Cap 2 - Potência e Raiz

Exemplos

Ex 2.1 $2^x \cdot 2^2$; $x = 3$

$2^{3+2} \Rightarrow 2^5 =$ **32**

Ex 2.2 $\frac{3^5}{3^2} = 3^{5-2} =$ **9**

Ex 2.3 a) $(4^3)^2 = 4^6 =$ **4096**

b) $5^{-1} = \frac{1}{5} =$ **0,2**

Ex 2.4 a) $(4x)^3 = 4^3 \cdot x^3 =$ **$64x^3$**

b) $(3\sqrt{x})^4 \Rightarrow 3^4 \cdot x^{4/2} \Rightarrow$

$81x^2$

Ex 2.5 a) $\left(\frac{2}{3}\right)^2 = \boxed{\frac{4}{9}}$

b) $\left(\frac{1}{5}\right)^2 = \boxed{\frac{1}{25}}$

Ex 2.6 a) $0,00473 \Rightarrow \boxed{4,73 \cdot 10^{-3}}$

b) $12345\,00000 \Rightarrow \boxed{1,23 \cdot 10^{9}}$

Ex 2.7 a) $\sqrt[5]{6^2} \Rightarrow \boxed{6^{2/5} \simeq 2,048}$

b) $\sqrt[3]{2^3} = 2^{3/3} \Rightarrow \boxed{2}$

Ex 2.8 $\sqrt[3]{a^3 \cdot b^6} = a^{3/3} \cdot b^{6/3} = \boxed{a \cdot b^2}$

Ex 2.9 $\sqrt{\frac{a^6}{b^5}} = \frac{a^{6/2}}{b^{5/2}} = \boxed{\frac{a^3}{\sqrt{b^5}}}$

Ex 2.10 $\sqrt[3]{\sqrt[2]{3}} = \sqrt[6]{3} = \boxed{3^{1/6}}$

Ex 2.11 a) $\sqrt{3} + 4\sqrt{3} - 2\sqrt{3} \Rightarrow (1+4-2)\sqrt{3}$

$\boxed{3\sqrt{3}}$

b) $2\sqrt[5]{3} + 3\sqrt[5]{3} - 2\sqrt[5]{3} \Rightarrow$

$\boxed{3\sqrt[5]{3}}$

c) $4\sqrt{2} - 2\sqrt{2} + 3\sqrt{5} - 6\sqrt{5} \Rightarrow$ $\boxed{2\sqrt{2} - 3\sqrt{5}}$

d) $3\sqrt{2} + 7 - 5\sqrt{2} - 4 + 5\sqrt{12} \Rightarrow$

$\boxed{-3\sqrt{2} + 5\sqrt{12} + 3}$

Ex. 2.12 a) $2\sqrt{2}\ 3\sqrt{5} \Rightarrow$ $\boxed{6\sqrt{10}}$

b) $\sqrt[3]{x.y} \cdot \sqrt[3]{x^2} \cdot \sqrt[3]{y^4} = \sqrt[3]{x^3 . y^5} = x^{3/3} . y^{5/3}$

$x \sqrt[3]{y^5} = x \sqrt[3]{y^3 . y^2} =$

$\boxed{xy\sqrt[3]{y^2}}$

(04)

c) $\sqrt{3} \cdot \sqrt[4]{2} = \sqrt[8]{3^4 \cdot 2^2} = \sqrt[8]{81 \cdot 4} = \boxed{\sqrt[8]{324}}$

d) $\dfrac{\sqrt{x^3}}{\sqrt{xy}} = \dfrac{\sqrt{x \cdot x^2}}{\sqrt{x} \cdot \sqrt{y}} = \dfrac{x\sqrt{x}}{\sqrt{x}\sqrt{y}} = \dfrac{x}{\sqrt{y}}$

Racionalizando o denominador

$$\frac{x}{\sqrt{y}} \cdot \frac{\sqrt{y}}{\sqrt{y}} = \boxed{\frac{x\sqrt{y}}{y}}$$

e) $\dfrac{\sqrt{2}}{\sqrt[3]{5}} \cdot \dfrac{\sqrt[3]{5^2}}{\sqrt[3]{5^2}} = \dfrac{\sqrt[6]{2^3 \cdot 5^4}}{5} = \boxed{\dfrac{\sqrt[6]{5000}}{5}}$

Lista de Exercícios

Potência e Raiz

(01) a) $0,002 = \boxed{2 \cdot 10^{-3}}$

b) $12000 \Rightarrow \boxed{1,2 \cdot 10^{4}}$

c) $0,213 \Rightarrow \boxed{2,13 \cdot 10^{-1}}$

d) $1,2 \cdot 10^{2} + 6 \cdot 10^{3} \Rightarrow 1,2 \cdot 10^{2} + 60 \cdot 10^{2}$

$(60 + 1,2) \cdot 10^{2} \Rightarrow 61,2 \cdot 10^{2} \Rightarrow \boxed{6,12 \cdot 10^{3}}$

(02) a) $\left(\dfrac{x^{2^4} \div (x^{2})^{3}}{x^{2^4} \cdot x^{-16}} \right)^{-2} \Rightarrow \left(\dfrac{x^{16} \cdot x^{-16}}{x^{16} \div x^{6}} \right)^{2}$

$\left(\dfrac{1}{x^{10}} \right)^{2} \Rightarrow \boxed{\dfrac{1}{x^{20}}}$

(02)

b) $\left(\frac{4^{-2}\cdot(-3)^{3}}{(-3)^{-1}}\right)^{-2} = \left(\frac{(-3)^{-1}}{4^{-2}\cdot(-3)^{3}}\right)^{2}$

$\frac{1/9}{4^{-4}\cdot 729} = \frac{1/9}{\frac{1}{256}\cdot 729} = \boxed{\frac{256}{6561}}$

(03) $(0,2)^{3} + (0,16)^{2} = \left(\frac{2}{10}\right)^{3} + \left(\frac{16}{100}\right)^{2}$

$\frac{8}{1000} + \frac{256}{10000} = \frac{80+256}{10000} = \frac{336}{10000}$

$\boxed{0,0336}$

(04) $\frac{2^{-1} - (-2)^{2} + (-2)^{-1}}{2^{3} + 2^{-2}} = \frac{-4}{4+\frac{1}{4}}$

$\frac{-4}{17/4} = \boxed{-\frac{16}{17}}$

(05) $\dfrac{\frac{3}{2} - 0{,}05}{\left(\frac{4}{5}\right)^{-1}} = \dfrac{\frac{3}{2} - \frac{5}{100}}{\left(\frac{5}{4}\right)}$

$\dfrac{30 - 1}{20} \cdot \dfrac{4}{5} = \boxed{\dfrac{29}{25}}$

(06) $10^{2x} = 25 \Rightarrow (10^x)^2 = 5^2$

Logo $\Rightarrow 10^x = 5^1$, então $10^{-x} = 5^{-1}$,

$\boxed{\frac{1}{5}}$

(07) $\dfrac{2^{n+4} - 2 \cdot 2^n}{2 \cdot 2^{n+3}} = \dfrac{2^4 \cdot 2^n - 2 \cdot 2^n}{2 \cdot 2^3 \cdot 2^n}$

$\dfrac{2^4 - 2}{2^4} = \dfrac{14}{16} = \boxed{\dfrac{7}{8}}$

(07)

(08) $\left(a^{-2} + b^{-2}\right)^{-1} = \dfrac{1}{\dfrac{1}{a^2} + \dfrac{1}{b^2}}$

$$\frac{1}{\dfrac{b^2 + a^2}{a^2 b^2}} = \boxed{\frac{(a \cdot b)^2}{a^2 + b^2}}$$

(09) $a = 2^3$; $b = a^2$; $c = 2^a$; $2 \cdot a \cdot b \cdot c = ?$

Então: $2 \cdot 2^3 \cdot \left(2^3\right)^2 \cdot 2^{2^3} = 2 \cdot 2^3 \cdot 2^6 \cdot 2^8$

$$\boxed{2^{17}}$$

(10) $8\sqrt{5} - \sqrt{20} + \sqrt{45} + 5\sqrt{180}$

$8\sqrt{5} - \sqrt{4}\sqrt{5} + \sqrt{9} \cdot \sqrt{5} + 5\sqrt{36}\sqrt{5}$

$8\sqrt{5} - 2\sqrt{5} + 3\sqrt{9} + 30\sqrt{5}$

$$\boxed{39\sqrt{5}}$$

(11) $\sqrt[3]{2\sqrt{5}} = \sqrt[3]{2 \cdot 5^{1/2}} = \sqrt[3]{2} \cdot \sqrt[3]{5^{1/2}}$

$\sqrt[3]{2} \cdot \sqrt[6]{5} = \sqrt[6]{2^2} \cdot \sqrt[6]{5} = \boxed{\sqrt[6]{20}}$

(12) a) $\frac{4}{\sqrt{3}} = \boxed{\frac{4\sqrt{3}}{3}}$

b) $\frac{7}{\sqrt[3]{7}} \cdot \frac{\sqrt[3]{7^2}}{\sqrt[3]{7^2}} = \frac{7}{7}\sqrt[3]{7^2} = \boxed{\sqrt[3]{49}}$

c) $\frac{4}{3\sqrt{2}} \cdot \frac{\sqrt{2}}{\sqrt{2}} = \frac{4\sqrt{2}}{6} = \boxed{\frac{2\sqrt{2}}{3}}$

d) $\frac{4}{5-\sqrt{3}} \cdot \frac{(5+\sqrt{3})}{(5+\sqrt{3})} = \frac{20+4\sqrt{3}}{25-3}$

$\frac{20+4\sqrt{3}}{22} = \boxed{\frac{10+2\sqrt{3}}{11}}$

06

(13) a) $(a+\sqrt{a})(a-\sqrt{a}) + (a+\sqrt{a})^2 - 2a\sqrt{a}$

$a^2 - a + a^2 + 2a\sqrt{a} + a - 2a\sqrt{a}$

$a^2 + a^2 =$

$\boxed{2a^2}$

b) $x = 1 + \sqrt{2}$; $x^2 - 2x + 1$

$(1+\sqrt{2})^2 - 2(1+\sqrt{2}) + 1$

$1 + 2\sqrt{2} + 2 - 2 - 2\sqrt{2} + 1$

$\boxed{2}$

(14) a) $\sqrt{18} = \sqrt{3^2 \cdot 2} =$

$\boxed{3\sqrt{2}}$

18	2
9	3
3	3
1	

b) $\sqrt{75} = \sqrt{5^2 \cdot 3} =$

$5\sqrt{3}$

75	5
15	5
3	3
1	

c) $\sqrt[5]{\frac{32}{243}} = \frac{\sqrt[5]{32}}{\sqrt[5]{243}} = \boxed{\frac{2}{3}}$

d) $\sqrt[4]{625} = \sqrt[4]{5^4} =$ **5**

625	5
125	5
25	5
5	5
1	

15) a) $a^{\frac{2}{3}} =$ $\sqrt[3]{a^2}$

b) $4^{\frac{1}{16}} =$ $\sqrt[16]{4}$

c) $16^{\frac{1}{2}} = \sqrt{16} =$ **4**

16) a) $\sqrt{0,25} = \sqrt{\frac{25}{100}} = \frac{\sqrt{25}}{\sqrt{100}} = \frac{5}{10} =$ **0,5**

b) $\sqrt[3]{64} = \sqrt[3]{2^3 \cdot 2^3} = 2 \cdot 2$

4

64	2
32	2
16	2
8	2
4	2
2	2
1	

2^3 2^3

(08)

c) $\sqrt[5]{243} = \sqrt[5]{3^5} =$ **3**

243	3
81	3
27	3
9	3
3	3
1	

3^5

d) $\sqrt{10^2 - 9^2} = \sqrt{100 - 81} =$ $\boxed{\sqrt{19}}$

(17) a) $\sqrt[3]{m^5} = \sqrt[3]{m^3 \cdot m^2} =$ $\boxed{m\sqrt[3]{m^2}}$

b) $\sqrt[4]{7^9} = \sqrt[4]{7^4 \cdot 7^4 \cdot 7} =$ $\boxed{49\sqrt[4]{7}}$

c) $\sqrt[3]{x^3 \cdot m^5} = \sqrt[3]{x^3 \cdot m^3 \cdot m^2} =$ $\boxed{xm\sqrt[3]{m^2}}$

d) $\sqrt{a^2 \cdot n^3} = \sqrt{a^2 \cdot n^2 \cdot n} =$

$\boxed{an\sqrt{n}}$

(18) a) $\frac{\sqrt{20}}{\sqrt{4}} = \sqrt{\frac{20}{4}} = \boxed{\sqrt{5}}$

b) $\frac{\sqrt{12}}{\sqrt[3]{5}} = \frac{\sqrt{2^2 \cdot 3}}{\sqrt[3]{5}} = \frac{2\sqrt{3}}{\sqrt[3]{5}} \cdot \frac{\sqrt[3]{5^2}}{\sqrt[3]{5^2}} = \frac{2\sqrt{3}\sqrt[3]{25}}{5}$

c) $\frac{\sqrt[5]{5}}{\sqrt[3]{10}} = \frac{\sqrt[5]{5} \cdot \sqrt[3]{10^2}}{10} = \boxed{\frac{5^{1/5} \cdot 10^{2/3}}{10}}$

d) $\sqrt{2} \cdot \sqrt{18} = \sqrt{36} = \boxed{6}$

(19) $\sqrt{13 + \sqrt{7 + \sqrt{2 + \sqrt{4}}}} = \sqrt{13 + \sqrt{7 + \sqrt{4}}}$

$\sqrt{13 + \sqrt{9}} = \sqrt{16} = \boxed{4}$

(20) $\sqrt{13^2 - 12^2} = \sqrt[n]{125}$

Aqui, podemos resolver de duas maneiras:

(1) Desenvolvendo os quadrados: $\sqrt{169 - 144} = \sqrt[n]{125}$

$\sqrt{25} = \sqrt[n]{125}$

$5 = \sqrt[n]{125}$

$\boxed{n = 3}$

(10)

(21) Utilizando a propriedade: $a^2 - [a-1]^2 = a + a - 1$

Logo: $13^2 - 12^2 = 25 \Rightarrow \sqrt{25} = \sqrt[n]{125}$

$5 = \sqrt[n]{125} = \boxed{n=3}$

(21) $\dfrac{(\sqrt{2}+\sqrt{3})}{\sqrt{3}} \cdot \dfrac{\sqrt{3}}{\sqrt{3}} \Rightarrow \boxed{\dfrac{\sqrt{6}+3}{3}}$

(22) $\dfrac{1+\sqrt{3}}{\sqrt{3}-1} \cdot \dfrac{(\sqrt{3}+1)}{\sqrt{3}+1} = \dfrac{1+2\sqrt{3}+3}{(\sqrt{3})^2 - 1^2}$

$\dfrac{4+2\sqrt{3}}{2} = \boxed{2+\sqrt{3}}$

(23) $\dfrac{7\sqrt{3} - 5\sqrt{48} + 2\sqrt{192}}{3\sqrt{3}}$

$\dfrac{7\sqrt{3} - 5\sqrt{2^2 \cdot 2^2 \cdot 3} + 2\sqrt{2^2 \cdot 2^2 \cdot 2^2 \cdot 3}}{3\sqrt{3}}$

$\dfrac{7\sqrt{3} - 20\sqrt{3} + 16\sqrt{3}}{3\sqrt{3}} = \dfrac{3\sqrt{3}}{3\sqrt{3}} = \boxed{1}$

(24) $\frac{\sqrt{50}-\sqrt{8}}{\sqrt{2}} \cdot \frac{\sqrt{2}}{\sqrt{2}} = \frac{\sqrt{100}-\sqrt{16}}{2}$

$\frac{10-4}{2} = \frac{6}{2} = \boxed{3}$

(25) $\frac{2-\sqrt{2}}{\sqrt{2}-1} \cdot \frac{\sqrt{2}+1}{\sqrt{2}+1} = \frac{2\sqrt{2}+2-2-\sqrt{2}}{2-1}$

$\boxed{\sqrt{2}}$

(26) a) $\frac{\sqrt[3]{16}}{\sqrt[3]{2}} = \sqrt[3]{\frac{16}{2}} = \sqrt[3]{8} = \boxed{2}$

b) $\frac{\sqrt{2}}{\sqrt[3]{6}} \cdot \frac{\sqrt[3]{6^2}}{\sqrt[3]{6^2}} = \boxed{\frac{\sqrt{2}\cdot\sqrt[3]{36}}{6}}$

c) $\sqrt[5]{8}\cdot\sqrt[5]{4} = \sqrt[5]{32} = \boxed{2}$

d) $\sqrt[3]{5}\cdot\sqrt{4} = \sqrt[6]{25\cdot 64} = \sqrt[6]{1600}$

$\sqrt[6]{1600} = \sqrt[6]{2^6\cdot 25} = \boxed{2\sqrt[6]{25}}$